CWE-Bench Task Project

Build a scalable pipeline that produces 100~200 Harbor-format security-fix tasks in 3 weeks. targeting CWE-Bench. CWE-Bench is private - there are no reference tasks, but the category is ***Security fixes: realistic vulnerabilities in real code paths***

Goal: every accepted task score pass@3 = 0 on both DeepSeek V4.1 Flash (max) and Xiaomi MiMo-V2.6-Pro.

A failure counts only if the task passes your well-defined task rubric check and trial analysis attributes it to model capability, not a task defect. Quality comes first.

Task spec

Each task pairs a vulnerable real repo with a realistic report. Hidden tests pass only if the fix closes the hole and keeps the feature working.

Component	**Spec**
Format	Harbor: `instruction.md`, `task.toml`, `environment/`, `solution/` (oracle), `tests/` (hidden)
Environment	Real open-source repo in its vulnerable state; dependencies and any required services (DB, cache) installed
Instruction	Realistic vulnerability report: observed impact and how it is triggered. No fix, no file or function hints
Security tests (fail → pass)	Exploit the vulnerability through a real entry point and assert it is blocked, including variants beyond any PoC in the report
Functional tests (pass → pass)	Existing tests of the affected code plus legitimate-use cases, so over-blocking or removing the feature fails
Grading	Partial score
Metadata	CWE ID, language, repo, source (mined or injected), entry point

Out of scope: build-from-scratch or secure code generation; detection-only or exploit-only tasks; fixes that are only a dependency bump or config change; contrived snippets or unreachable code; LLM as judge